

Flutter Environment Setup Guide

1. Install Required Software

Install Git

Download and install Git first:

[Git Official Website](#)

After installation, verify:

```
git --version
```

2. Install Android Studio

Download Android Studio:

[Android Studio Official Website](#)

During installation make sure these are selected:

- Android SDK
 - Android SDK Platform
 - Android Virtual Device (AVD)
-

3. Install Flutter SDK

Official installation guide:

[Flutter Install Docs](#)

Download Flutter SDK:

- Extract ZIP file
- Example path:
 - `C:\flutter`

4. Add Flutter to PATH

Windows

1. Search: "Environment Variables"
2. Edit System Environment Variables
3. Open Environment Variables
4. Edit **Path**
5. Add:

C:\flutter\bin

Verify:

flutter --version

Configure Android Studio for Flutter

Install Flutter & Dart Plugins

Open Android Studio:

- Settings
- Plugins
- Search:
 - Flutter
 - Dart

Install both and restart Android Studio.

Setup Android Emulator

Create AVD

1. Open Android Studio

2. More Actions → Virtual Device Manager
3. Create Device
4. Select:
 - Pixel device
5. Download recommended Android image
6. Finish setup

Start emulator.

Verify Flutter Installation

Run:

```
flutter doctor
```

Expected:

- Flutter ✓
- Android toolchain ✓
- Chrome ✓
- Android Studio ✓
- Connected device ✓

Fix any red errors.

Create Your First Flutter App

Command

```
flutter create first_app
```

Move into project:

```
cd first_app
```

Run app:

```
flutter run
```

Understand Flutter Project Structure

Important Folders

lib/

Contains Dart code.

Main entry point:

lib/main.dart

android/

Native Android files.

ios/

Native iOS files.

pubspec.yaml

Project dependencies and assets.

Example:

dependencies:

flutter:

 sdk: flutter

Install dependencies:

flutter pub get

build/

Generated compiled files.

Learn Flutter CLI Commands

Create Project

`flutter create app_name`

Run App

`flutter run`

Build APK

`flutter build apk`

Install Dependencies

`flutter pub get`

Clean Project

`flutter clean`

Check Connected Devices

`flutter devices`

Check Environment

`flutter doctor`

Hot Reload vs Hot Restart

Hot Reload

- Keeps app state
- Fast UI updates

Shortcut:

r

Hot Restart

- Restarts app state

Shortcut:

R

Recommended YouTube Channels

- Flutter Official
- Traversy Media
- Reso Coder
- The Flutter Way

Useful searches:

- “Flutter installation complete guide”
 - “Android Studio setup for Flutter”
 - “Flutter emulator setup”
 - “Flutter project structure explained”
 - “Flutter first app tutorial”
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GitHub Repository Setup

Create repository:

[GitHub](#)

Initialize Git:

```
git init  
git add .  
git commit -m "Initial Flutter setup"
```

Connect repository:

```
git remote add origin YOUR_GITHUB_REPO_URL  
git push -u origin main
```

Final Deliverables Checklist

- ✓ Android Studio installed
- ✓ Flutter SDK configured
- ✓ Emulator working
- ✓ First Flutter app running
- ✓ `flutter doctor` all checks passed
- ✓ Screenshot captured
- ✓ README created
- ✓ GitHub repository uploaded